



A Preliminary Proposal for a Systematic GABC Encoding of Gregorian Chant

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ABSTRACT

In the last years, several approaches have addressed the encoding of the different music scripts used for plainchant. One of these approaches is the GABC format. While being a comprehensive symbolic representation of square notation, the lack of a formal specification for GABC usually leads to ambiguities, which must be avoided in the specification of any encoding format. Sometimes, the simple trial-and-error approach of entering the GABC code into an engraving system—such as Illuminare, Scrib.io, or GABC Transcription Tool—can solve this ambiguity. However, these engraving systems have shown some inconsistency among themselves when rendering GABC, sometimes displaying different music for the same code snippet. This paper presents a systematic approach to encoding Gregorian chant originally written in Aquitanian neumes and square notation to eliminate ambiguities inherent in the GABC specification. By formalizing the grammar of GABC, we address the challenges of inaccurate renderings in current music notation software. Our methodology includes developing a “Systematic GABC” (S-GABC) following a critical and scientific mentality to ensure the endurance of the notation. This paper demonstrates our system’s effectiveness in standardizing Gregorian chant encoding, offering significant contributions to digital musicology and enhancing the accuracy of musical heritage digitization.

CCS CONCEPTS

• **Applied computing** → **Document scripting languages**; **Document management and text processing**; **Sound and music computing**; • **Information systems** → **Information retrieval**; **Data encoding and canonicalization**.

KEYWORDS

Plainchant, square notation, Aquitanian neumes, Gregorian chant, GABC, music encoding, MEI



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1 INTRODUCTION

This paper is the result of the work of two projects, ECHOES¹ and Repertorium,² that use GABC³ as an intermediate format to encode plainchant originally written in Aquitanian neumes and square notation. This encoding is then converted into Neumes MEI, an XML-based format that encodes chant sources in their original music script [3, 8, 9].⁴ For both projects, obtaining MEI Neumes files as the final output of their encoding process is important as MEI will be used for music information retrieval purposes (in the case of the Repertorium project) and for computational music analysis (in the ECHOES project).⁵ Furthermore, this format is gaining acceptance in the scholar community due to its rich and well-structured information encoding. However, generating MEI from scratch proves too intricate for our project workflows. Since GABC is a typesetting (text-based) format, it is easier to encode chants in GABC—by manually typing it or by using it as the semantic representation in an optical music recognition (OMR) workflow—and then to convert the GABC encodings into Neumes MEI than it is to encode the chants directly into MEI.

The two projects handle the GABC to MEI conversion process differently as they have different goals. ECHOES project provides a conversion script, and for the Repertorium project a grammar-driven process is used. Regardless of the approach, both research groups have noticed inconsistencies between the GABC encodings, their renderings by different software, and the documentation. This paper summarizes the findings of both research groups, presenting a preliminary approach to encoding Gregorian chant that eliminates

¹<https://echoes.fcsh.unl.pt>

²<https://repertorium.eu>

³<https://gregorio-project.github.io/gabc/>

⁴<https://music-encoding.org/guidelines/v5/content/neumes.html>

⁵Although other formats exist for encoding chants (like volpiano [4, 10] and MonodiPlus [1]), MEI is the most comprehensive one in terms of capturing both the pitch and the shape of the neume components (whether it is a punctum, a virga, an ornamental note like a quilisma, oriscus, or liquescent, etc).

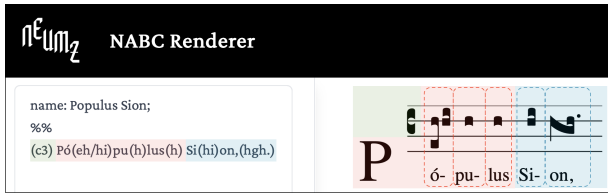


Figure 1: Example of the GABC encoding and rendering of the beginning of a chant using Scrib.io, with the clef (in green) and the words “Pópulus” (in red) and “Sion” (in blue). Full example at <https://gregorio-project.github.io/gabc/details.html>.

the ambiguities inherent in the GABC specification by: (1) formalizing the grammar of GABC to address the challenges of inaccurate renderings in current music notation software, and (2) proposing new encodings for features that currently are not supported by the format. Our methodology includes developing a Systematic GABC (S-GABC) following a critical, systematic, and scientific mentality to ensure that the notation is fully and accurately captured in the encoding.

The paper starts with an introduction to GABC in Section 2, followed by a presentation of the main GABC symbols used in both previously mentioned projects in Section 3. In Section 4, we present the S-GABC grammar. We discuss the inconsistencies found across the GABC engraving applications in Section 5, and present a summary of unsupported features in GABC and our proposed solution in Section 6. In Section 7, we report on the equivalences of GABC with MEI. Finally, we draw a summary of the paper and conclusions in Section 8.

2 BRIEF INTRODUCTION TO GABC

GABC is a typesetting language used to encode plainchant. It was created by a monk from the Abbey of Sainte Madeleine du Barroux for the purpose of Gregorian chant publications following the example of books published by the Abbey of Solesmes.⁶ GABC can encode both the text and music content of Gregorian chants as shown in the incipit provided in Figure 1. This figure displays two words: “Pópulus” and “Sion,” highlighted in red and blue, respectively. In the encoding, shown on the left side of Figure 1, the two words are separated by a space character “ ”. The syllables of each word contain the neumes associated with them within parentheses. The height of the individual notes in these neumes (i.e., their location on the staff) is given by the characters *a* to *m*, following the guidelines shown in Figure 2. In GABC, all the music content is enclosed within parentheses. For example, the C3 clef is encoded as (c3), and the neumes associated to their respective syllables are placed between a pair of parentheses following the syllable’s text (like the two-note rising movement expressed by hi, belonging to the syllable “Si” encoded as Si(hi)). Neume boundaries within a syllable are expressed in GABC using a slash, see Po(eh/hi).

In addition to the GABC letters *a* to *m* to indicate the location of a note on the staff, GABC uses certain characters as prefixes or suffixes to these letters to provide information about the neumes



Figure 2: Relation between GABC letters and staff position (from <https://gregorio-project.github.io/gabc/index.html>).

and the various graphical shapes they take (as described in the next section).

3 EVALUATION OF EXISTING GABC SYMBOLS

Tables 1 and 2 provide a summary of the neume shapes and word-syllable-neume structure supported by GABC. The tables do not include the full set of features supported by GABC as they exclude those like separation bars, accents and attachments, text formatting and special characters, less common neume spacing, custodes, and polyphony features. In the ECHOES project, we did not use these features. Regarding spacing, we used only two characters: the slash / character to signal the neume boundaries within a syllable—the GABC “neumatic cut”—specified in the grammar rule (gr.18), and the regular space character to separate the words in the chant—the GABC “word space”; both of them are included in Tables 1 and 2. In ECHOES, we excluded the other less standard space-related characters; namely, the @ character to indicate a connection between notes (gr.17), the exclamation mark ! (gr.16) to indicate no connection between the notes, the double slash // to indicate large neumatic cuts, the half-space /0 character, and the mix of slash with bracketed numbers /[factor] to scale the neumatic cut. While these more sophisticated space-related features are useful in the typesetting of chants to indicate how big or small a space is, they were deemed unnecessary for our purposes of recording the boundaries between neumes, syllables, and words in a chant.

4 SYSTEMATIC GABC (S-GABC) GRAMMAR

Two main documentation sources can be found for encoding GABC, the official one maintained by the Gregorio Project,⁷ and a complementary collaborative wiki page.⁸ Despite the tutorials and the comprehensive list of examples, being written in plain text, they lack the formality necessary to avoid ambiguities. For this reason, we propose an EBNF grammar that specifies any possible encoding situation found in both above mentioned projects. As many of these encoding situations are not clearly described in the different documentation sources, and we have made some encoding decisions, we denote this grammar Systematic-GABC to be a contribution towards a systematic approach to encoding Gregorian chant with GABC.

The syntax rules in Figure 3, along with the lexical grammar described in Table 3, formally specify the GABC used in the ECHOES and Repertorium projects so far. We have removed some symbols for the sake of the readability of the paper, which can be consulted in a full version in the GitHub repository.⁹ Note that this grammar

⁶According to Matthias Bry, founder and head of the Nocturnale Romanum project, originally, GABC was based on Solesmes books published or re-edited after 1980, such as Psalterium Monasticum [5] or Liber hymnarius [6].

⁷<http://gregorio-project.github.io/gabc/index.html>

⁸<https://gregoriochant.org/>

⁹<https://github.com/repertorium/gabc2mei>

Feature	GABC Encoding	Example
CLEF		
C and F clefs, from lines 1 to 5 (never exceeding the number of staff lines given in the header) (gr.10)	<i>c</i> 1 to <i>c</i> 5 and <i>f</i> 1 to <i>f</i> 5	See Figure 1: (<i>c</i> 3)
NOTE HEIGHT AND NOTE HEAD SHAPE		
Staff location (square note head) (gr.22)	lowercase characters from <i>a</i> to <i>m</i>	<i>g</i>
Staff location (rhombus) (gr.23)	uppercase characters from <i>A</i> to <i>M</i>	<i>G</i>
SUFFIXES: stems		
Downward stem (Right side of note) (gr.27)	<i>v</i>	<i>gv</i>
Downward stem (Left side of note) (gr.26)	<i>V</i>	<i>gV</i>
PREFIXES: stems		
Removes stem at the beginning of the neume (gr.17)	@	<i>gf</i> <i>@gf</i> <i>gfh</i> <i>@gfh</i>
SUFFIXES: ornaments (note shape)		
Liquescent (gr.32)	~	<i>g ~</i>
	(Rising melody)	<i>gh ~</i>
	(Falling melody)	<i>gh ~</i>
Liquescent (two tails down) (gr.33)	>	<i>g ></i>
Liquescent (two tails up) (gr.34)	<	<i>g <</i>
Oriscus (gr.31)	<i>o</i>	<i>go</i>
Quilisma (gr.30)	<i>w</i>	<i>gw</i>
Strophicus (gr.29)	<i>s</i>	
	(Distropha)	<i>gs/gs</i> or <i>gss</i>
	(Tristropha)	<i>gs/gs/gs</i> or <i>gsss</i>

Table 1: Summary of GABC main symbols. The corresponding grammar rules are specified between parentheses.

SUFFIXES: rhythmic sign (to be encoded after note shape signs)			
Punctum mora (gr.43)	.	<i>g.</i>	
Vertical episema or ictus (gr.40)	,	<i>g'</i>	
Horizontal episema (gr.41)	-	<i>g_</i>	
SUFFIXES: accidentals			
Flat (gr.36)	<i>x</i>	<i>gx</i>	
Natural (gr.37)	<i>y</i>	<i>gy</i>	
Sharp (gr.38)	#	<i>g#</i>	
BOUNDARIES			
Separation of neumes within a syllable (gr.18)	regular neumatic cut: /	See Figure 1: <i>Po(eh/hi)</i>	
Syllables (gr.8)	parentheses: ()	See Figure 1:	
Words (gr.1)	spaces	<i>Po()pu()lus() Si()on,()</i>	

Table 2: Summary of GABC main symbols (cont). The corresponding grammar rules are specified between parentheses.

is designed to consider its later ANTLR4 parser generation implementation [7], which first parses the lexical tokens to be used later in the syntactic parser. This forces us to specify many lexical rules in the syntax rules in a non-compact way.

5 INCONSISTENCIES BETWEEN ENGRAVING SOFTWARES

At the moment, there are three different online platforms and one downloadable program for rendering GABC: Scrib.io (also called NABC Renderer),¹⁰ Source & Summit Editor (the Alpha version of the earlier Illuminare Editor),¹¹ GABC Transcription Tool,¹² and the Gregorio desktop app.¹³ After testing these four applications, we noticed some inconsistencies in the display of certain features between the rendering software, which should be addressed.¹⁴ Table 4 presents a summary of these features.

The Gregorio desktop application has some advantages to the other renders as shown in Table 4: it is possible to transcribe on a 5-line staff and allows a C clef on the 5th line (Features 1 and 2), which is something that GABC supports according to its documentation.¹⁵

¹⁰<https://scrib.io>

¹¹<https://www.sourceandsummit.com/editor/alpha/>

or <https://gregorio.illuminarepublications.com>

¹²<https://bbloomf.github.io/jgabc/transcriber.html>

¹³To be downloaded at <https://run.gregoriochant.org/cgi/process.pl>

¹⁴For this work, we have discarded GregorioTeX and other engraving systems oriented to obtaining a printing-ready score sheet instead of a consistent music encoding, for which usually ad-hoc layout commands are used as patches for ambiguous situations, such as the `\greseteolcustos` instruction used in GregorioTeX for switching off the automatic custos engraving.

¹⁵See the mention to "staff-lines" in the "Header" section and the last bullet point of "The clef" section in the Gregorio documentation (<https://gregorio-project.github.io/gabc/>).

TOKEN	DEFINITION						
char_a	'a'	char_v	'v'	minus	'-'	open_lyric	'<sp>'
char_b	'b'	char_w	'w'	num_0	'0'	open_bold	''
char_c	'c'	char_x	'x'	num_1	'1'	open_italic	'<i>'
char_d	'd'	char_z	'z'	num_2	'2'	open_color	'<c>'
char_e	'e'	char_O	'O'	num_3	'3'	open_underline	'<u>'
char_f	'f'	char_V	'V'	num_4	'4'	open_small_caps	'<sc>'
char_g	'g'	char_W	'W'	num_5	'5'	close_lyric	'</sp>'
char_h	'h'	char_Z	'Z'	num_6	'6'	close_bold	''
char_i	'i'	char_A2M	[A-M]	question_mark	'?'	close_italic	'</i>'
char_j	'j'	char_lt	'<'	exclam_mark	'!'	close_color	'</c>'
char_k	'k'	char_gt	'>'	slash	'/'	close_underline	'</u>'
char_l	'l'	char_tilde	'~'	at	'@'	close_small_caps	'</sc>'
char_m	'm'	char_hash	'#'	sq_bracket_open	'['	v_lyric	'V/'
char_o	'o'	dot	'.'	sq_bracket_close	']'	r_lyric	'R/'
char_s	's'	apostrophe	''	parenth_open	'('	a_lyric	'A/'
		underscore	'_'	parenth_close	)'	oe_lyric	'ôe'
						ae_lyric	'âe'

Table 3: Lexical grammar.

In case of a C5 clef, it would be helpful to be able to write a few notes above the one represented by the *m* character (which lies on the space above the fifth line). The documentation does not mention that this is possible. However, the characters *n* and *p* can be used to achieve this in Scrib.io and Gregorio, but are not supported by the other two editors (see Feature 3). Rendering two clefs in the staff, which GABC supports according to the documentation, is only possible using the Sources & Summit Editor or the Gregorio desktop app (see Feature 4). These two, together with the GABC Transcription Tool, are also able to render a B-flat next to the clef, another feature supported by GABC according to its documentation.

Another inconsistency with the rendering systems has to do with the '@' and '!' characters. The characters '@' (gr.17) and '!' (gr.16) affect glyph determination, as they connect and disconnect notes and affect the groupings of them within a neume. However, the effect of these characters on glyph determination varies between the different rendering software. We tested over fifty-four combinations of these two signs ('@' and '!') in a four-note neume. The Gregorio Desktop app and Scrib.io were almost identical in their results (see blue cells in Table 4), with only three encoding samples (out of the fifty-four in total) resulting in different renderings. These three used the '@' character and are included in Features 8 and 9 in Table 4. The Source & Summit Editor and the GABC Transcription Tool also showed nearly identical results among themselves (see pink cells in Table 4). These two tools in particular (Source & Summit Editor and GABC Transcription Tool) are less susceptible to changes in the glyph shape when using a '@' character when compared to the other two tools (Gregorio Desktop and Scrib.io), as exemplified by Feature 7 of Table 4 (and also shown in Features 8 and 9). Around 33% of the studied neume engravings were different between these two pairs of GABC typesetting applications—the pair of Source & Summit Editor and GABC Transcription Tool (pink cells in the table) vs. the pair of Gregorio Desktop app and Scrib.io (blue cells in the table). An example of this is shown by Feature 6 in Table 4, where stems disappear in Source & Summit Editor and GABC Transcription Tool (pink cells).

In all GABC renderers, the same four-note neume shape (with or without stem, joined or separated in different places) can be obtained through different encodings using the signs '!' and '@'. As a result, sometimes it is unclear for the user to understand the exact function of a sign like '!', which is probably the most

complex because of the richness of outcomes. Meanwhile, other musical figures can be obtained only through a specific encoding (for example, in Scrib.io, a descending and then ascending musical gesture as four square notes with a downwards stem to the left of the first note: *hgf@g*, as shown in Feature 8 of Table 4; and a variant of the earlier without the stem: *@hgf@g*).

Lastly, the documentation includes information about ornamental neumes like quilisma and oriscus, encoded with the lowercase characters 'w' and 'o', respectively. However, it includes some seemingly random examples with these characters in uppercase letters, with no explanation of the distinct scenarios that would require their lowercase or uppercase use. In the "Gregorio and Gregorio-TEX" three hundred-page manual, there is some clarification about the uppercase and lowercase characters for quilisma and oriscus—with the 'O' being used for *oriscus scapus*, which is an oriscus with stems that connect to the note prior to it; and 'W' for *quilisima quadratum*, which is a quilisma with a distinct square quilisma shape [2, p.258]. However, the only system that fully respects these distinctions is the Gregorio Desktop app (see the last column of the rendering systems for features 10–13 and all other cells in green).

6 MISSING FEATURES AND PROPOSAL

In addition to the inconsistencies described above in section 5, during our projects, we noticed some missing features in GABC. These features and our proposed solutions are detailed below.

6.1 Encoding of initial stems

In the ECHOES project, we used the 'V' (gr.26) and 'v' (gr.27) characters to explicitly indicate a downward stem placed either to the left or the right side of a neume, regardless of the behaviour of the GABC rendering systems—this is, regardless of whether these rendering engines would automatically draw a stem or not by just encoding the notes' heights (i.e., pitches). This choice allows us to retain some important graphical features that are particularly relevant for early music, which often shows a variety of shapes that are unparalleled in modern notation. Because we explicitly encoded the initial stem of a neume, we did not use the '@' character to indicate the absence of the initial stem of a neume—as the mere absence of the *V* character would indicate that already. Moreover, the '@' character has more than one use (e.g., removing the initial stem in

#	Feature	Code Example	Scrib.io	Source & Summit Editor	GABC Transcription Tool	Gregorio
1	Five-line staff (the default is a four-line staff)	staff-lines:5	No	No	No	
2	C5 clef	(c5)	No	No	No	
3	Characters <i>n</i> and <i>p</i> for notes in the sixth staff line and the above space, respectively. This is not included in any documentation material (with characters going from <i>a</i> to <i>m</i>).	m n p		No	No	
4	Rendering of two clefs at the same time	(c2@c4)	No		No	
5	Rendering of a B-flat in a C clef	(cb3)	No			
6	Inconsistency in stems	ih!gf ih!gf	 	 	 	 
7	Susceptibility to @	fghi f@ghi fg@hi fgh@i	   	   	   	   
8	Use of @ in a ligature (descending+ascending)	hgf@g				
9	Use of @ in ligature (descending)	ih@gf ihg@f	 	 	 	 
Use of upper- and lowercase for W and O for ornamental neumes						
10	Oriscus (lowercase o)	egoi				
11	Oriscus (uppercase O)	egOi				
12	Quilisma (lowercase w)	gwi				
13	Quilisma (uppercase W)	gWi				

Table 4: Inconsistencies Between Rendering Software

```

(gr.1) body → syllable+
(gr.2) syllable → (syl_lyric_symbols syllable?
    close_text_modifiers? syl_musical_symbols?)
(gr.3) syl_lyric_symbols → lyric_symbols
    open_text_modifiers?
(gr.4) lyric_symbols → open_lyric lyric close_lyric
(gr.5) lyric → v_lyric | r_lyric | a_lyric | oe_lyric | ae_lyric
(gr.6) open_text_modifiers → open_bold | open_italic
    | open_color | open_underline | open_small_caps
(gr.7) close_text_modifiers → close_bold | close_italic
    | close_color | close_underline | close_small_caps
(gr.8) syl_musical_symbols → (parenth_open
    musical_symbol* parenth_close)
(gr.9) musical_symbol → clef | note | custos
(gr.10) clef → clef_symbol clef_flat? clef_number
(gr.11) clef_symbol → char_c | char_f
(gr.12) clef_flat → char_b
(gr.13) clef_number → num_1 | num_2 | num_3 | num_4
    | num_5
(gr.14) note → prefix pitch suffix* attachment*
(gr.15) prefix → no_space | remove_first_stem
    | neumatic_cut
(gr.16) no_space → exclam_mark
(gr.17) remove_first_stem → at
(gr.18) neumatic_cut → slash (slash | space_factor?)
(gr.19) space_factor → num_0 | sq_bracket_open minus?
    factor_number sq_bracket_close
(gr.20) factor_number → num_1 | num_2 | num_3 | num_4
    | num_5 | num_6
(gr.21) pitch → square_pitch | rhombus_pitch
(gr.22) square_pitch → 'a' .. 'm'
(gr.23) rhombus_pitch → char_A2M
(gr.24) suffix → stem | shape | accidental | rhythmic_sign
(gr.25) stem → virga_leftt | virga_right
(gr.26) virga_left → char_V
(gr.27) virga_right → char_v
(gr.28) shape → strophicus | quilisma | oriscus
    | liquescent | liquescent_two_tails_down
    | liquescent_two_tails_up
(gr.29) strophicus → char_s
(gr.30) quilisma → char_w | char_W
(gr.31) oriscus → char_o | char_O
(gr.32) liquescent → char_tilde
(gr.33) liquescent_two_tails_down → char_gt
(gr.34) liquescent_two_tails_up → char_lt
(gr.35) accidental → flat | natural | sharp
(gr.36) flat → char_x
(gr.37) natural → char_y
(gr.38) sharp → char_hash
(gr.39) rhythmic_sign → episema_vertical
    | episema_horizontal | punctum_mora
(gr.40) episema_vertical → apostrophe
(gr.41) episema_horizontal → underscore
    position_horizontal_episema?
(gr.42) position_horizontal_episema → num_0 | num_1
    | num_2 | num_3 | num_4 | num_5
(gr.43) punctum_mora → dot
(gr.44) custos → char_z num_0? | char_Z

```

Figure 3: GABC grammar

a neume, indicating a connection between notes, and having two clefs in one staff). Therefore, to avoid ambiguity, we did not include this character in the ECHOES project’s GABC encodings of chants. Instead, we propose to remove rule (gr.17) from the grammar.

6.2 Oblique shape

It is not possible to indicate which two neume components within a neume should be displayed as an oblique shape rather than two individual squares (see blue box in Figure 4), and vice versa (see green box in Figure 4). As seen in Figure 4, the rendering software makes an internal decision about the oblique part of a neume, which cannot be changed. Therefore, a character to explicitly indicate the beginning of an oblique section—always made of two neume components in a downward motion—is necessary. We suggest using a ‘o’ character preceding the first neume component of the two that form the oblique shape to clarify the use of oblique ligatures; when the character is not used, the assumption will be that the notes are square-shaped.

6.3 Clarifying lines

These are vertical or diagonal lines added by a second hand to clarify which notes belong to different syllables in ambiguous cases. These marks were added when the place of the syllable change is unclear; for example, when music is not vertically aligned with its text. Another example is shown in Figure 5, where the line breaks the ligature into two parts, indicating that the first of the two notes belongs to the preceding syllable “fi”, and the second note belongs to the syllable “li”.

6.4 Different liquescent shapes

There are many types of liquescent shapes in chant sources (Figure 6 shows a few of them), and GABC is not fully capable of capturing all of them. Because of this, in the ECHOES project, we made our own rules to account for the existence of augmentative and diminutive liquescent, and the different shapes they take in Aquitanian script. This is shown in the tables of Section 7 and the ECHOES project documentation.¹⁶

6.5 Marking of uncertain readings

Occasionally, the material condition of the early music document is poor, and it does not allow a clear identification of the melodic reading. In those cases, we suggest using the empty-note character ‘r’ from GABC to encode this. Empty notes are not used in plainchant (only in polyphony). Therefore, we can safely utilize the ‘r’ character to represent the neumes of an uncertain reading for Gregorian chant. Similarly, lacunas (completely illegible readings) can be encoded by leaving the content of the parentheses (i.e., the neumes of a syllable) empty.

The proposed grammar (shown in Figure 3) formally details the different lexical symbols used across the different encoding cases described with syntactic rules. This formalism helps us to propose new symbols for ambiguous or unsupported GABC cases (like those mentioned above and summarized in Table 5) safely, without overwriting any other cases.

¹⁶See the GABC to MEI conversion table at <https://github.com/ECHOES-from-the-Past/GABCToMEI/>

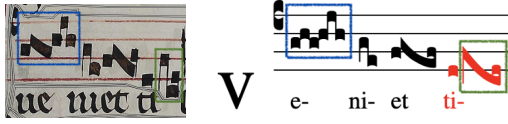


Figure 4: Example of ligatures from the Antiphonarium Benedictum, Cod. St. Georgen 6, 5v.



Figure 5: Added line to clarify or change the grouping of notes. Bibliothèque National de France, Bibliothèque de l'Arsenal 197 f. 10r.

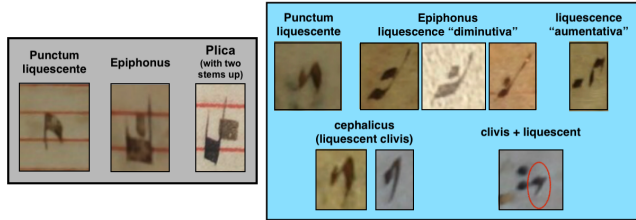


Figure 6: A few examples of liquescent shapes in square (gray box) and Aquitanian (blue box) music scripts.

#	Feature	Proposed Symbol	Code Example
1	Oblique shape	°	°hfh
2	Clarifying lines	"	hv"gih
3	Uncertain reading	r	

Table 5: Summary of proposed missing symbols

7 MEI ENCODING

As mentioned above, the ECHOES and Repertorium projects, use GABC as an intermediary encoding and export it to MEI for other tasks (e.g., music analysis). MEI and GABC use different approaches to encoding plainchant. MEI uses a more graphical point of view to avoid the different possible interpretations of the elements in the manuscripts and focuses on the musical content itself, while GABC focuses on a more typographical point of view. Tables 6, 7, and 8 show the equivalences between both formats for the main encoding features.

8 SUMMARY AND CONCLUSIONS

This paper presented a formal approach to encoding Gregorian chant by developing a Systematic GABC (S-GABC), addressing the notable ambiguities in the GABC specification and the missing features in the format. By formalizing the grammar and systematically addressing encoding decisions, including introducing supplementary symbols and explicit encoding rules, this work aims to develop

Feature	Example	MEI Encoding
NOTE SHAPES		
Staff location (square note head) (gr.22)	<i>g</i>	<nc/>
Rhombus (gr.23)	<i>G</i>	<nc tilt="se"/>
Downward stem (Right side of note) (gr.27)	<i>gv</i>	<nc tilt="s"/>
Downward stem (Left side of note) (gr.26)	<i>gV</i>	<nc tilt="n"/> (with tilt="ne" for Aquitanian)
Liquescent (gr.32)	<i>g ~</i>	<nc> <liquescent/> </nc> Depending on the melodic contour of the neume, add a @curve attribute (see below)
(Rising melody)	<i>gh ~</i>	<nc curve="a"/> <nc> <liquescent/> </nc>
(Rising melody with stem V)	<i>ghV ~</i>	<nc> <nc tilt="ne" curve="c"/> <liquescent/> </nc> (Only happens in Aquitanian)
(Falling melody)	<i>hg ~</i>	<nc> <nc curve="c"> <liquescent/> </nc>
Liquescent (two tails down) (gr.33)	<i>g ></i>	<nc curve="c"> <liquescent/> </nc>
Liquescent (two tails up) (gr.34)	<i>g <</i>	<nc curve="a"> <liquescent/> </nc>
Oriscus (gr.31)	<i>go</i>	<nc> <oriscus/> </nc>
Quilisma (gr.30)	<i>gw</i>	<nc> <quilisma/> </nc>

Table 6: GABC - MEI equivalences

Strophicus (gr.29)	Distropha: <i>gs/gs</i> or  <i>gss</i>	<code><nc></code> <code><strophicus/></code> <code></nc></code> <code><nc></code> <code><strophicus/></code> <code></nc></code>
	Tristropha: <i>gs/gs/gs</i> or <i>gsss</i> 	<code><nc></code> <code><strophicus/></code> <code></nc></code> <code><nc></code> <code><strophicus/></code> <code></nc></code> <code><nc></code> <code><strophicus/></code> <code></nc></code>
RHYTHMIC SIGNS (always to be encoded after the note-shape signs)		
Punctum mora (gr.43)	<i>g.</i> 	<code><nc dots="1"/></code>
	<i>g..</i> 	<code><nc dots="2"/></code>
Vertical episema (ictus) (gr.40)	<i>g'</i> 	<code><nc></code> <code><episema form="v"/></code> <code></nc></code>
Horizontal episema (gr.41)	<i>g_</i> 	<code><nc></code> <code><episema form="h"/></code> <code></nc></code>
ACCIDENTALS		
Flat (gr.36)	<i>gx</i> 	<code><accid accid="f"/></code> The location of the accidental in the staff (provided by a @loc attribute) is given by the GABC letter (<i>a-m</i>) that precedes the GABC accidental (<i>x, y, #</i>).
Natural (gr.37)	<i>gy</i> 	<code><accid accid="n"/></code> See previous row's note about accidentals.
Sharp (gr.38)	<i>g#</i> 	<code><accid accid="s"/></code> See previous row's note about accidentals.

Table 7: GABC - MEI equivalences (cont.)

a more robust version of GABC to contribute to digital musicology and the preservation, encoding, and accessibility of musical heritage.

This work, which serves as technical documentation for the ECHOES and Repertorium projects, underscores the dynamic nature of S-GABC, anticipating its evolution as further ambiguities in the format's specification are discovered and solved. The development of a comprehensive grammar and rule set provides a clear

BOUNDARIES AND SPACING		
Separation of neumes within a syllable	See Figure 1: <i>Po(eh/hi)</i>	<code><syllable></code> <code><syl>Po</syl></code> <code><neume></code> <code><nc loc="2"/></code> <code><nc loc="5"/></code> <code></neume></code> <code><neume></code> <code><nc loc="5"/></code> <code><nc loc="6"/></code> <code></neume></code> <code></syllable></code>
Syllables (gr.8)	See Figure 1: <i>Po()pu()lus() Si()on, ()</i>	<code><syllable></code> <code><syl wordpos="i">Po</syl></code> <code><neume/></code> <code></syllable></code> <code><syllable></code> <code><syl wordpos="m">pu</syl></code> <code><neume/></code> <code></syllable></code> <code><syllable></code> <code><syl wordpos="t">lus</syl></code> <code><neume/></code> <code></syllable></code>
Words (gr.1)		<code><syllable></code> <code><syl wordpos="i">Si</syl></code> <code><neume/></code> <code></syllable></code> <code><syllable></code> <code><syl wordpos="t">on,</syl></code> <code><neume/></code> <code></syllable></code> The start and end of a word is given by its word position @wordpos in <syl> (which can be initial "i", middle "m", terminal "t", or single-syllable word "s").

Table 8: GABC - MEI equivalences (cont.)

roadmap for processing Gregorian chant, paving the way for future case studies and applications that will further validate and refine the system.

In summary, this nascent systematic format marks a critical step forward in standardizing the format for the accurate encoding of Gregorian chant, which will evolve as more ambiguities in the GABC specifications are found. This evolution will be reflected in future works, providing studies on how S-GABC has been applied in the ECHOES and Repertorium projects.

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